

PretrainLab-X

Day 8 Final Technical Report

Auditable, engineering-scale mini foundation-model pretraining stack

Date: 2026-09-14. This report summarizes checked-in evidence. It does not claim industrial-scale pretraining or a causal effect of data freshness alone.

Key results

Evidence	Result	Boundary
Strict Day 2-3	30.42M params; 500 steps	byte-level fallback corpus
Day 4 benchmarks	336M/653M short runs	no 3B/5B-token long run
Day 5	336M; 500,006,912 positions; loss 0.069339	9.9M slice replayed
Day 6	336M; 500,006,912 unique tokens	single-GPU bounded run
Shared held-out	NLL 8.424211 vs 3.039400; Δ 5.384811	not a causal proof
Day 7	load/config/random/precision checks pass	no weight edits

1. System

The model is a pre-normalized decoder-only Transformer using RMSNorm, RoPE, SwiGLU and GQA (16 query heads / 4 key-value heads). Causal attention uses PyTorch SDPA with the recorded `sdpa_auto` backend. The trainer implements BF16 autocast, accumulation, clipping, warmup-cosine scheduling, atomic checkpoints and complete optimizer/scheduler/RNG restoration.

2. Data

The real-data path uses FineWeb-Edu sample-10BT and the TinyLlama tokenizer. Day 4 retains a 10M-token slice; Day 6 uses a separate shard and document-boundary split. Raw Parquet, token arrays, checkpoints and W&B; binaries remain private.

3. Interpretation

Day 5 replays a 9.9M-token slice while Day 6 consumes a fresh stream. Their shared held-out difference is stable under paired bootstrap and sanity checks, but the histories are not identical, so the result is not a freshness-only causal estimate.

4. Audit

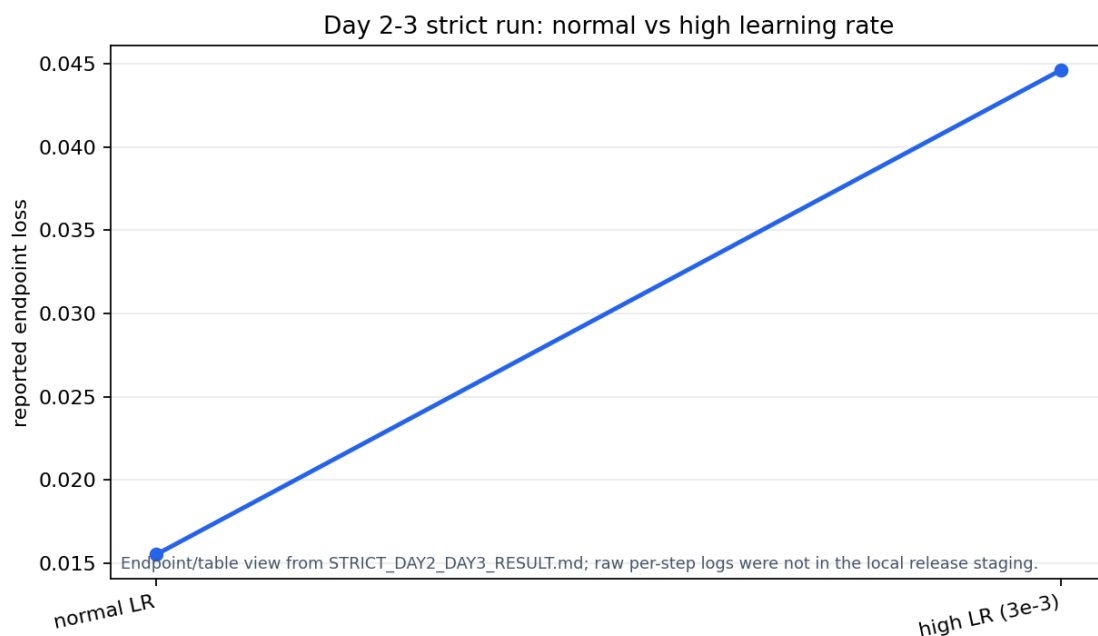
`results/final_metrics.json` is generated from checked-in summaries. `audit/` contains the canonical SHA-256 manifest and cross-check. `scripts/verify_release.py` rejects forbidden large or credential-like artifacts.

5. Reproduction

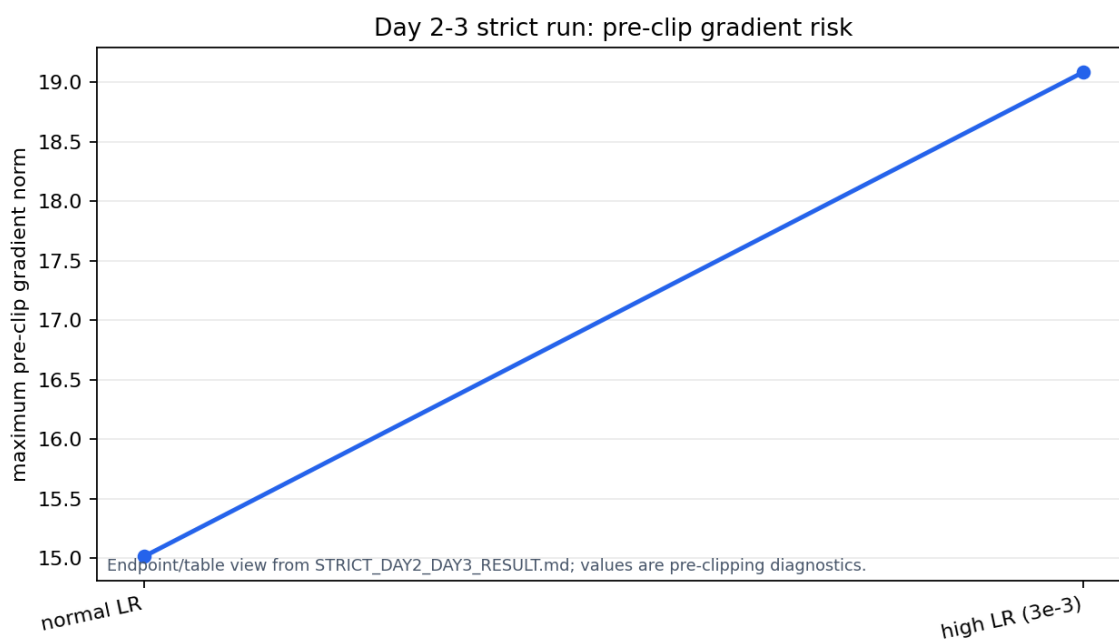
Start with the CPU smoke test in `REPRODUCIBILITY.md`, then use the explicitly named short GPU benchmarks. Long-run reproduction requires private inputs and storage; no public command downloads third-party raw data automatically.

Figures

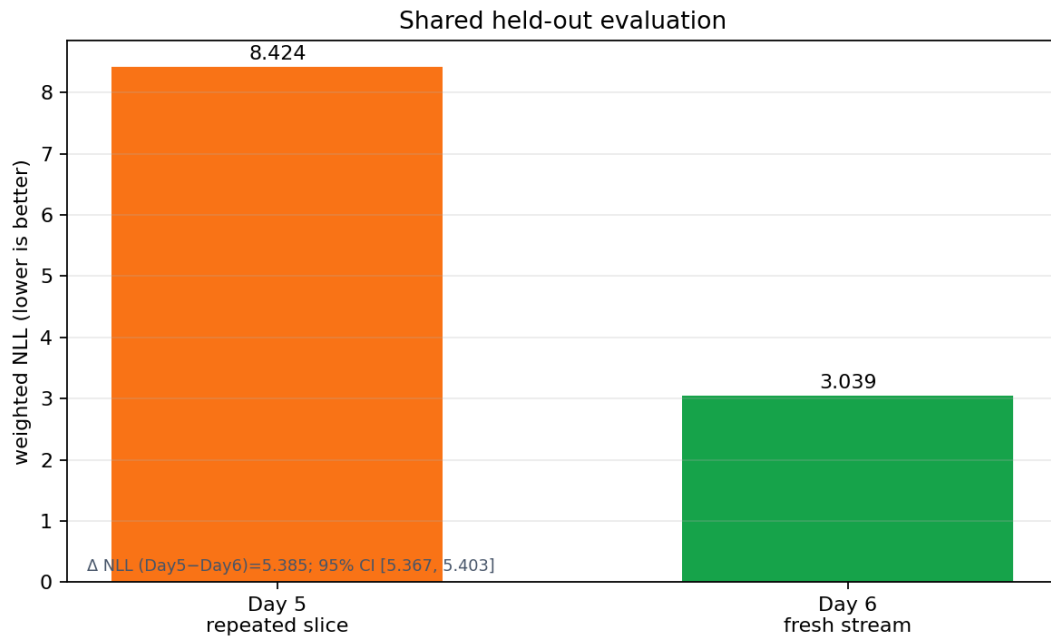
Strict Day 2-3 endpoint loss comparison



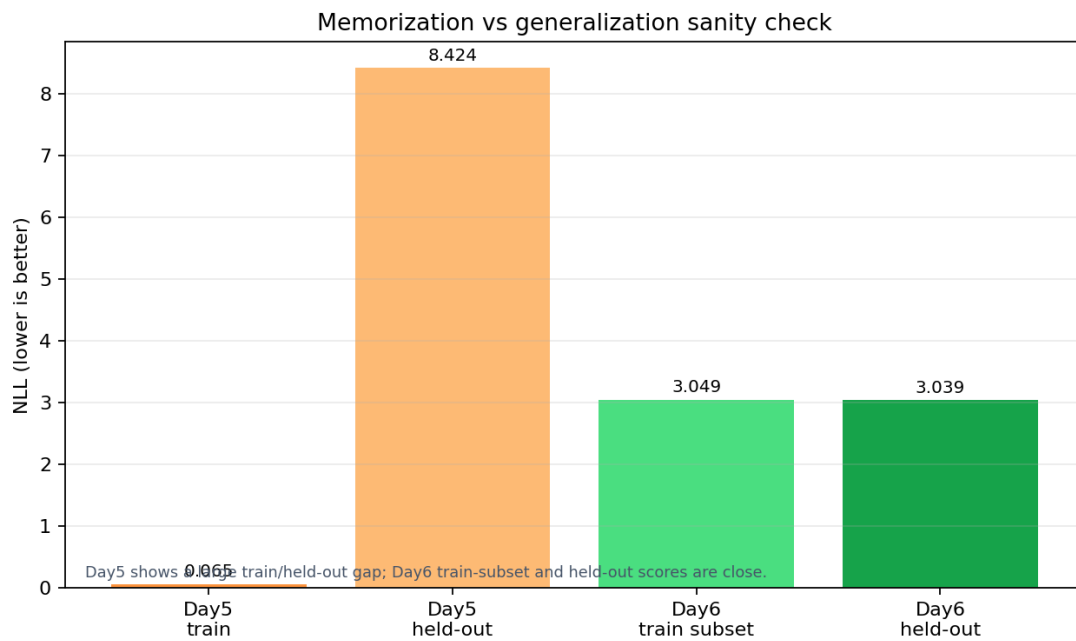
Strict Day 2-3 pre-clip gradient risk



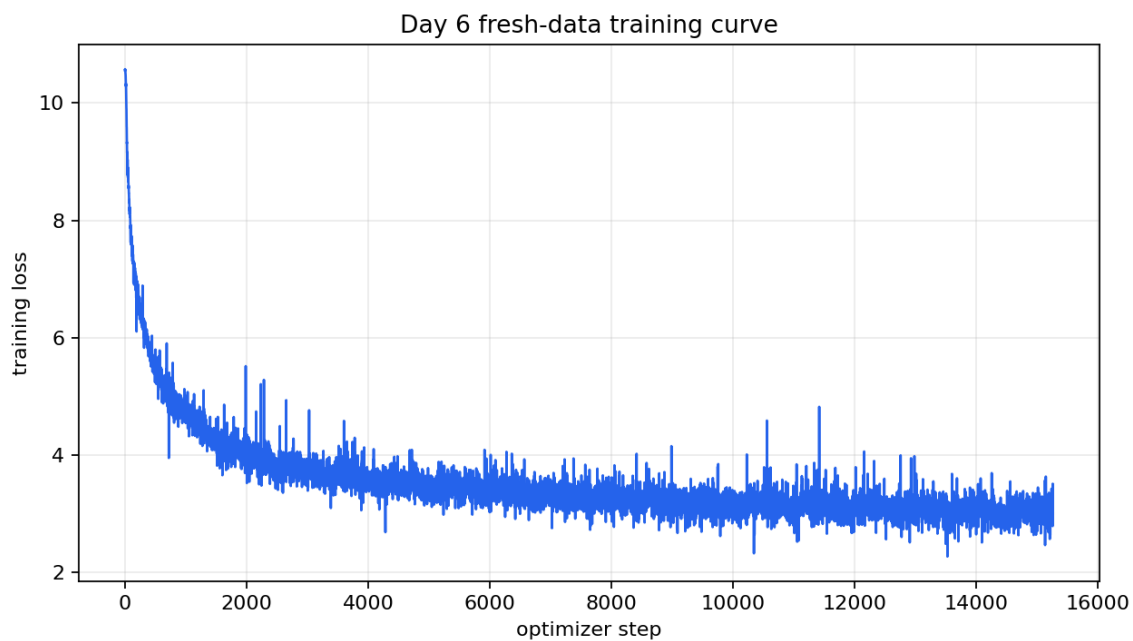
Shared held-out NLL



Memorization versus generalization



Day 6 fresh-data training curve



See README.md, README_CN.md and LIMITATIONS.md for the public release boundary. Values and source paths are machine-readable in results/final_metrics.json.